

	SGGS Institute of Engineering and Technology, Nanded	
	Computer Science & Engineering	
	In-Semester Term Examination 2025 (Semester-I)	
	Big Data Analysis (PEC-CS415)	B Tech Final Year (2025)
Date: 15/10/2025	Time: 14:30 to 15:15	Max. Marks: 20

Q. 1	Explain data storage in HDFS.	5	CO2 BT3
Q. 2	What is the purpose of the Job Tracker and task tracker in Hadoop	5	CO2 BT3
Q. 3	Explain Support Vector Machine in detail?	5	CO4 BT4 &5
Q. 4	Explain Decision Tree Algorithm?	5	CO4 BT4 &5

(a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.

(b) List all of the strong association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers and item: denotes variables representing items



Department of Computer Science and Engineering

Semester I (2025-26)

In Semester Examination -2

Date: 15/10/2025

Time: 02:30 to 03:15

Class: Final Year (CSE)

Maximum Marks: 20

Code: PEC-CS 408

Subject: (Elective III) Data Mining.

- Notes: 1. Figures to the right indicate full marks.
2. Solve any Two from the following.
3. Assume suitable data if necessary.

- (1) A database has 5 transactions. Let min sup = 60% and min conf = 80%.

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TID	items bought
T100	{ M, O, N, K, E, Y }
T200	{ D, O, N, K, E, Y }
T300	{ M, A, K, E }
T400	{ M, U, C, K, Y }
T500	{ C, O, O, K, I, E }

- (a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.
(b) List all of the strong association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers, and $item_i$ denotes variables representing items (e.g., "A", "B", etc.):
 $\forall x \in \text{transaction}, \text{buys}(X, \text{item1}) \wedge \text{buys}(X, \text{item2}) \Rightarrow \text{buys}(X, \text{item3}) [s, c]$.

- (2) Suppose that a data warehouse for Big-University consists of the following four dimensions: student, course, semester, and instructor, and two measures *count* and *avg grade*. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the *avg grade* measure stores the actual course grade of the student. At higher conceptual levels, *avg grade* stores the average grade for the given combination.

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- (a) Draw a snowflake schema diagram for the data warehouse.
(b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big-University student.

- (3.A) Briefly compare the following concepts. use an appropriate example to explain following points.

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- (a) Snowflake schema, fact constellation, starlet query model.
(b) Data cleaning, data transformation, refresh.

- (3.B) Describe and explain the different OLAP operations with appropriate examples.

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List out the differences between the OLAP and OLTP Systems.

Course Outcomes and B.T Level information



Q.No	B.T Level	C.O. satisfied
1	BT- 3	CO 4
2	BT- 2	CO 3
3	BT- 1	CO 2, CO 3

After successful completion of this course, student will be able to

CO1: Understanding the concept and Importance of data mining.

CO2: Students will learn to organize the data required and prepare it for mining using preprocessing techniques.

CO3: Perform exploratory analysis of the data to be used for mining.

CO4: Understanding the appropriate data mining methods like classification, clustering, or Frequent Pattern mining.

CO5: Mining data streams, social networks and big data.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE-2 Exam (Semester-I) 2025-26

Final Year B. Tech.

Computer Science and Engineering

Advance Database Management Systems

PCC-CS401

Date:14-10-2025 Time: 14.30 to 15.15

Max. Marks: 20 Marks

- Note:
- Attempt all questions.
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.
 - Write point to point answers in short.

Q. N	Question	Marks	CO	BT
Q.1	Solve any three questions. (3 x 4=12 M)	12 M		
a	Draw and Explain Tree-like interconnection architecture.	4 M	CO1	BT2
b	Analyze and Explain layers of distributed query processing with diagram.	4 M	CO2	BT4
c	For the given SQL query write down two equivalent relational algebra queries that are correct transformations of the query. SELECT ENAME FROM EMP, ASG WHERE EMP.ENO = ASG.ENO AND RESP = "Manager"	4 M	CO3	BT3
d	Explain xml with examples.	4 M	CO5	BT2
Q.2	Solve any two questions. (2 x 4 = 8 M)	8 M		
a	Consider an engineering firm that has offices in Boston, Waterloo, Paris and San Francisco. They run projects at each of these sites and would like to maintain a database of their employees, the projects and other related data. Assuming that the database is relational, we can store this information in two relations: EMP (ENO, ENAME, TITLE)1 and PROJ (PNO, PNAME, BUDGET). We also introduce a third relation to store salary information: SAL (TITLE, AMT) and a fourth relation ASG which indicates which employees have been assigned to which projects for what duration with what responsibility's (ENO, PNO, RESP, DUR). Write down the SQL query to find out the names and employees who worked on a project for more than 24 months.	4 M	CO2	BT2
b	Explain four characteristics of distributed query processor.	4 M	CO2	BT2
c	How does query execution in a Distributed Database Management System (DDBMS) differ from that in a Centralized Database Management System (CDBMS)?	4 M	CO2	BT4

Course Outcomes (CO)

- CO1: Apply concepts of distributed and parallel databases to design efficient database architectures and query processing strategies.
- CO2: Analyze transaction management techniques, including commit protocols, concurrency control, and fault tolerance in distributed and multidata base systems.
- CO3: Develop and evaluate advanced querying and information retrieval methods such as OLAP, data mining, and data warehousing for decision support systems.
- CO4: Demonstrate proficiency in handling advanced data types (spatial, multimedia, temporal, mobile) and their applications in modern database systems.
- CO5: Design and implement applications using emerging technologies such as XML databases, multidimensional indexing, and web-based database interfaces.
- CO6: Critically evaluate performance issues, tuning methods, and administration challenges in advanced DBMS for real-time, large-scale, and heterogeneous environments.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create